Challenge – Case Study – Solution

We are going to design a measurement system of the acoustic noise in wind tunnels tests based on an 8 microphones array (figure 1.1). Each microphone is a MEMS type piezo-electric sensor with two outputs; each one is conditioned with a low noise amplifier (LNA). One sensor section detects in the range from 100 Hz to 10 kHz and the other section detects in the range from 10 kHz to 100 kHz. Two analog multiplexers (see the blocks *switches*) concentrate the outputs of 8 microphones in two cables connected to an acquisition system remotely.

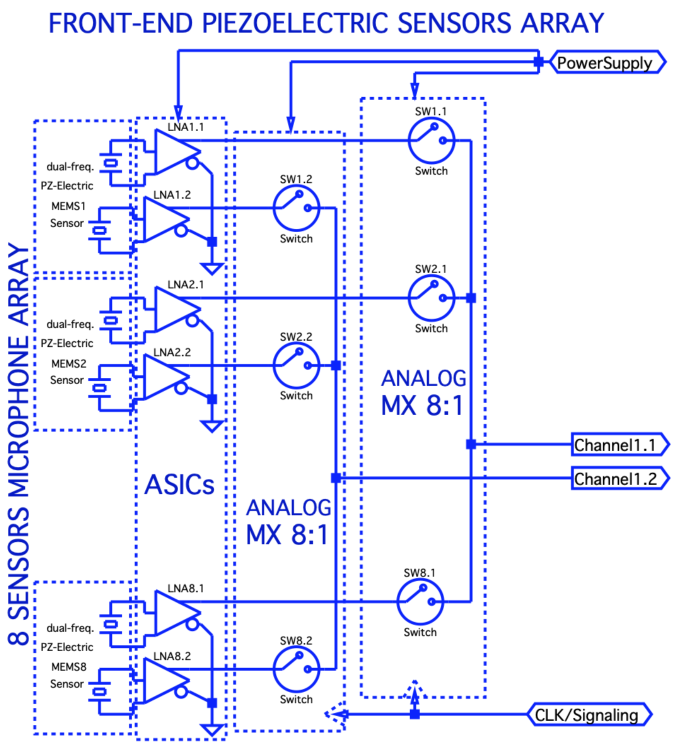

Figure 1.1 Bloc diagram of the microphone array

MEMS Sensor

Sensitivity 5 mV/Pa

Noise Equivalent Pressure 2 μPa/√Hz

LNA Amplifier

Gain 20 V/V

Bandwidth 100 kHz

Noise input voltage 1.5 nV/√Hz

Gain Drift 10 ppm/°C

AGC (Automatic gain control)

Gain between $-G_V$ and G_V (dB)

DAQ (acquisition system)

An A/D converter per channel

14 bits

input range ±1V

sampling from 30 MSps to 50 MSps

The characteristics in the wind tunnel tests (WTT) are the following: detection band of frequencies from 100 Hz to 100 kHz, temperature from -20°C to 70°C, input range from 30 dB_{SPL}[(1)] to 170 dB_{SPL},

[(1)] SPL: sound pressure level in units of dB relative to $2\cdot10^{-5}$ Pa RMS

1) Calculate the noise limited resolution in each section of the sensor.
2) Calculate the measurement error due to the thermal drift of the LNA.
3) Calculate the minimum gain of the AGC required to detect with the DAQ the acoustic signal in the worst case (input 30 dB_{SPL}). Calculate the sensitivity in dB re 1 V/Pa for this case.
4) Calculate the gain/attenuation of the AGC that is necessary to detect within the range of the DAQ the acoustic signal in the worst (input 170 dB_{SPL}). Give a value to the parameter G_V.
5) Choose a sampling frequency for each of two DAQ channels. Which is the minimum value of the clock frequency CLK that you propose for each multiplexer? Delay between channels?
6) Calculate the SNR (input 120 dB_{SPL}). Compare it with the target of 90 dB SNR.

1) Calculate the noise limited resolution in each section of the sensor.

   Noise N = $(NEP)^2 \cdot S^2 \cdot G^2 \cdot BW + (v_{ni})^2 \cdot G^2 \cdot BW$

   NEP (Noise equivalent pressure) vni (Noise input voltage LNA)

   S sensitivity G gain of LNA BW noise bandwidth = 100 kHz (output LNA)

   $$\text{Resol} = \sqrt{BW \cdot \left[NEP^2 + \left(\frac{v_{ni}}{S}\right)^2\right]} = \sqrt{100kHz \cdot \left[\frac{(2\mu Pa)^2}{Hz} + \left(\frac{1{,}5nV}{\frac{5nV}{\mu Pa}}\right)^2 \cdot \frac{1}{Hz}\right]} = 640\mu Pa$$

   Min: $30dB_{SPL} = 20log\frac{p}{20\ \mu Pa} \Rightarrow p_{min} = 633\ \mu Pa$

   Note that this approximates the worst case: The sensor has a limited BW of 9900 Hz for one section and 90 kHz for the other section. The noise bandwidth of the LNA is 100 kHz.

2) Calculate the measurement error due to the thermal drift of the LNA.

   $\Delta T = 70°C - (-20°C) = 90°C$ Relative error: $\frac{\Delta Vo}{Vo} = \frac{\Delta G}{G} = 20\frac{ppm}{°C} \cdot 90°C = 0.9\,^{\circ}/_{\circ\circ}$

   FS: $170dB_{SPL} = 20log\frac{p}{20\ \mu Pa} \Rightarrow p_{max} = 6.32\ kPa$ Error FE: $\Delta p_{\varepsilon} = 5.7\ Pa$

3) Calculate the minimum gain of the AGC that is necessary to detect with the DAQ the acoustic signal in the worst case (input 30 dB_{SPL}). Calculate the sensitivity in dB re 1 V/Pa for this case.

   Resol DAQ $= \frac{1V-(-1V)}{2^{14}-1} = 122\ \mu V$ $p_{min} = 633\ \mu Pa$

   $$Vo = G_{CAG} \cdot G_{LNA} \cdot S \cdot 633\ \mu Pa = G_{CAG} \cdot 20V/V \cdot 5\ \mu V/mPa \cdot 0.63\ mPa = 122\ \mu V$$

   $$G_{CAG} = 1.94\ V/V\ = 6\ dB$$

   Total sensitivity is $1.94 \cdot 20 \cdot 5\ \mu V/mPa\ = 0.194\ V/Pa\ = -14.24\ \text{dB re 1V/Pa}$

4) Calculate the gain/attenuation of the AGC that is necessary to detect within the range of the DAQ the acoustic signal in the worst (input 170 dB_{SPL}). Give a value to the parameter G_V.

   Span DAQ $= 2\ V$ $p_{max} = 6.32\ kPa$

   $$Vo = G_{CAG} \cdot G_{LNA} \cdot S \cdot 6.32\ kPa = G_{CAG} \cdot 20\ V/V \cdot 5\ V/kPa \cdot 6.32\ kPa = 2\ V$$

   $G_{CAG} = \frac{1}{316}V/V = -50\ dB$ $G_V = 50$

5) Choose a sampling frequency for each of two DAQ channels. Which is the minimum value of the clock frequency CLK that you propose for each multiplexer? Delay between channels?

   It is enough with the minimum possible sampling frequency 30 MSps

   To take about 20 samples per period:

   Section up to 10 kHz: 8 channels·10 kHz/channel·20 samples /period = 1.6 MHz (CLK)

   Section up to 100 kHz: 8 channels ·100 kHz/channel ·20 samples/period = 16 MHz (CLK)

   A sample rate multiple of the CLK would be 32 MSps

The delay between the first sampled channel and the eight sampled channel depends on the sampling rate:

a) *If the CLK is 32 MHz and the sampling rate is 32 MSps synchronized with the CLK, the maximum delay is (8-1) / 32 μs = 219 ns. In this case, both sections are oversampled.*

b) *If the CLK of a section is 1.6 MHz and the sampling rate is 32 MSps, 32 / 1.6 = 20 samples per channel are recorded continuously and the maximum delay is (8-1) / 1.6 μs = 4.375 μs. The 20 consecutive samples can be post-processed (averaging, denoising) to obtain a single sample, which is sufficient to reconstruct a 10 kHz signal.*

c) *If the CLK of the other section is 16 MHz and the sampling frequency is 32 MSps, 32 / 16 = 2 samples per channel are continuously recorded and the maximum delay is (8-1) / 16 μs = 437.5 ns.*

In wind tunnel test microphone array applications, a typical specification for the maximum relative delay is 1 μs. Case b) does not meet this specification. Either the CLK must be 8 MHz (delay 7 / 8 μs = 875 ns), or a sample and hold (S&H) stage must be included to sample all eight channels at the same instant of time. The specifications for the S&H are delay spread less than 1 μs and a hold time greater than 8 / 1.6 μs = 5 μs to allow the ADC to acquire all channels.

6) Calculate the SNR (input 120 dB_{SPL}). Compare it with the target of 90 dB SNR.
The noise is approximately 30 dB_{SPL} (question 1). The SNR is 120 dB_{SPL} - 30 dB_{SPL} = 90 dB.